Project title

AI-Assisted Grain Tracking and Quantitative Analysis of In Situ TEM analysis

In situ TEM experiments generate time-resolved videos showing the evolution of a large number of grains. At present, quantitative analysis of these videos is mainly performed manually or semi-manually. A typical video can contain hundreds of grains and many consecutive frames. It is therefore very time-consuming to manually identify and follow individual grains and quantify their evolution.

We would like to automate this analysis in order to extract quantitative information about grain dynamics, including:

- grain size and changes in grain size;
- grain shape and changes in grain shape;
- grain position and movement/displacement;
- growth, shrinkage, merging or disappearance of grains;
- statistical evolution of the grain population.

The main challenge is to automatically identify and track individual grains throughout the TEM video, rather than analysing only individual selected grains.

We would like to develop a Python-based image analysis program for automated or semi-automated analysis of in situ TEM videos.
The program should be able to:

1. identify/segment individual grains in TEM images;
2. track grains between consecutive video frames;
3. calculate quantitative parameters for individual grains, such as size, shape and position;
4. determine how these parameters change with time;
5. calculate movement/displacement of individual grains;
6. provide statistical analysis of the grain population.

The initial objective is a working proof-of-concept demonstrated on representative in situ TEM videos. A semi-automated workflow, where the user can correct segmentation or tracking errors when necessary, is acceptable. If appropriate, available machine-learning/image-segmentation methods may be evaluated as part of the development.